Automated Re-Training Pipeline Triggered by Data Drift

Assignment 8 | Demand Forecasting | MLOps PBA2

Student Details

Name:	Tanmay Shinde
Subject:	MLOps PBA2
Enrollment No.:	ADT23SOCB1617
Class:	LY-SMAD
GitHub:	github.com/tanmays0/ADT23SOCB1617-drift-retraining-pipeline

Problem Statement

Domain: Demand Forecasting
Tech Stack: Prefect or Airflow, Evidently AI, Scikit-Learn, MLflow

Build an automated pipeline that continuously monitors incoming data streams for data and concept drift using Evidently AI. When drift exceeds a defined threshold (e.g., Wasserstein distance > 0.1), automatically trigger a re-training DAG.

Key Tasks:
1. Generate synthetic drifted data.
2. Register newly trained models into an MLflow Registry.
3. Set up alert notifications (Email).

Tech Stack (Implemented)

Prefect | Evidently AI | Scikit-Learn | MLflow | Email alerts (local files)

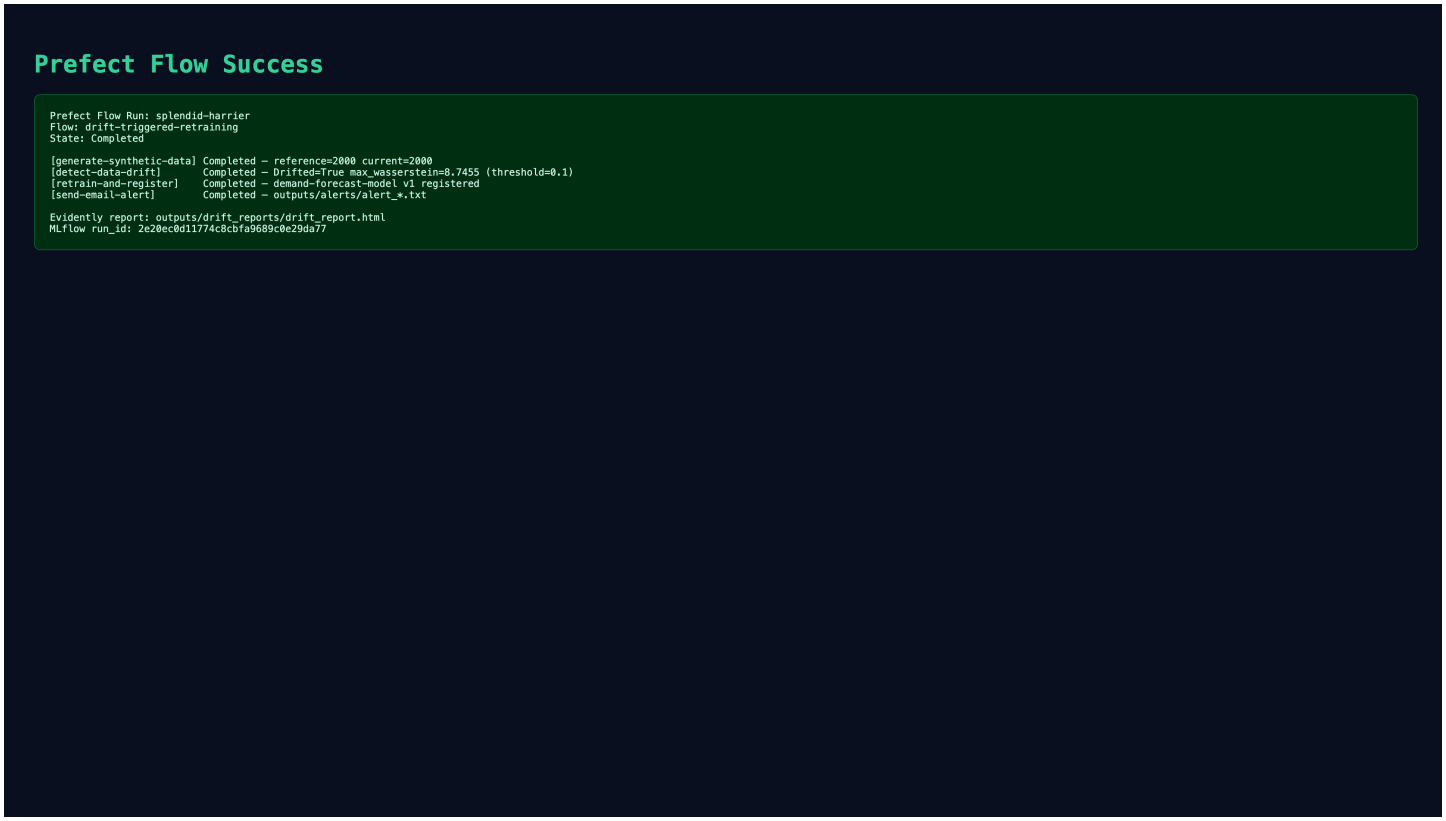
Key Results

- Drift detected: True (4/6 columns; share 0.667)
- Wasserstein (units_sold): 0.817 | price: 3.04 | temperature: 1.54
- Model registered: demand-forecast-model v1 (trigger=data_drift)
- Metrics: MAE ~ 3.52, R2 ~ 0.95
- Alert: email notification written to outputs/alerts/

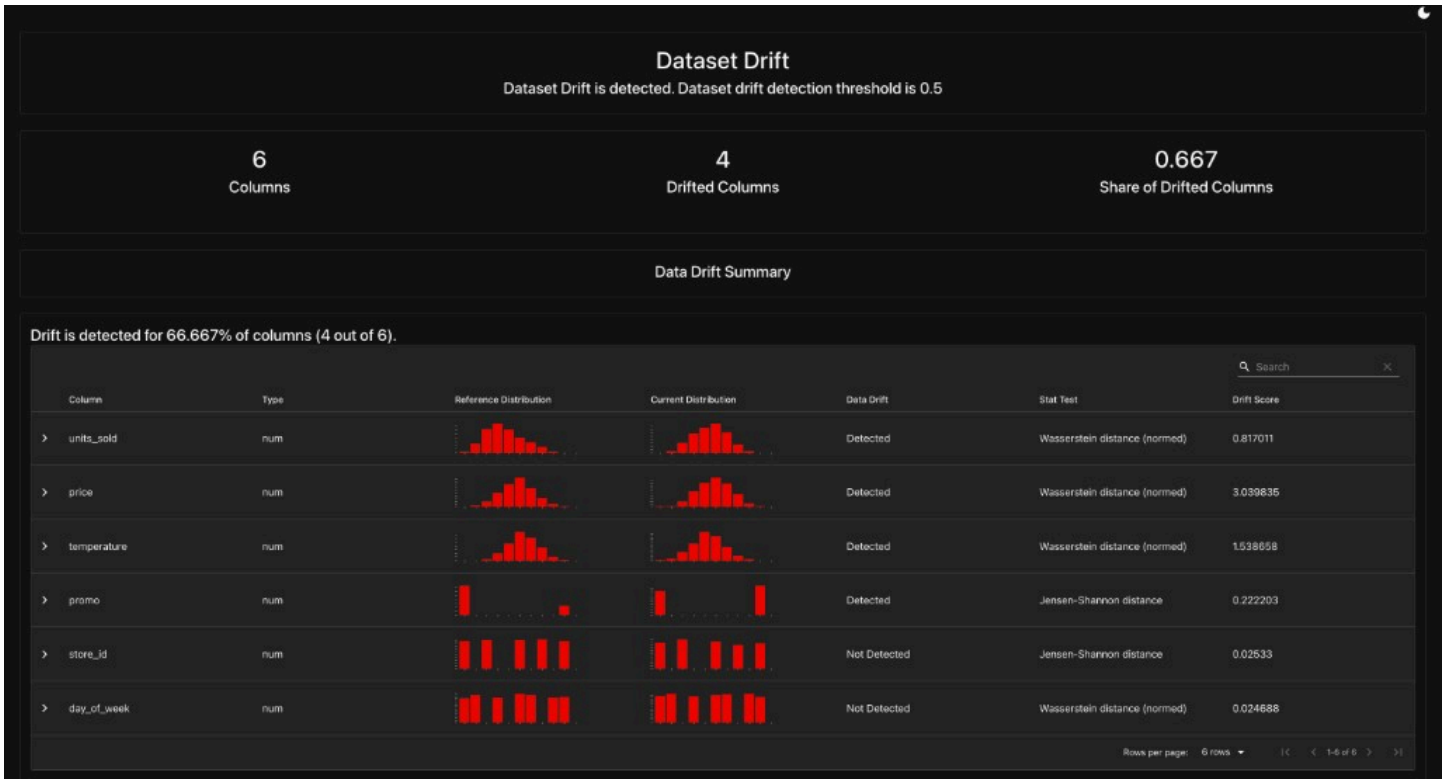
Where to check live output

GitHub: https://github.com/tanmays0/ADT23SOCB1617-drift-retraining-pipeline
MLflow UI: http://127.0.0.1:5000/#/models
Evidently report: outputs/drift_reports/drift_report.html
Email alert: outputs/alerts/alert_*.txt
Re-run: python scripts/run_demo.py

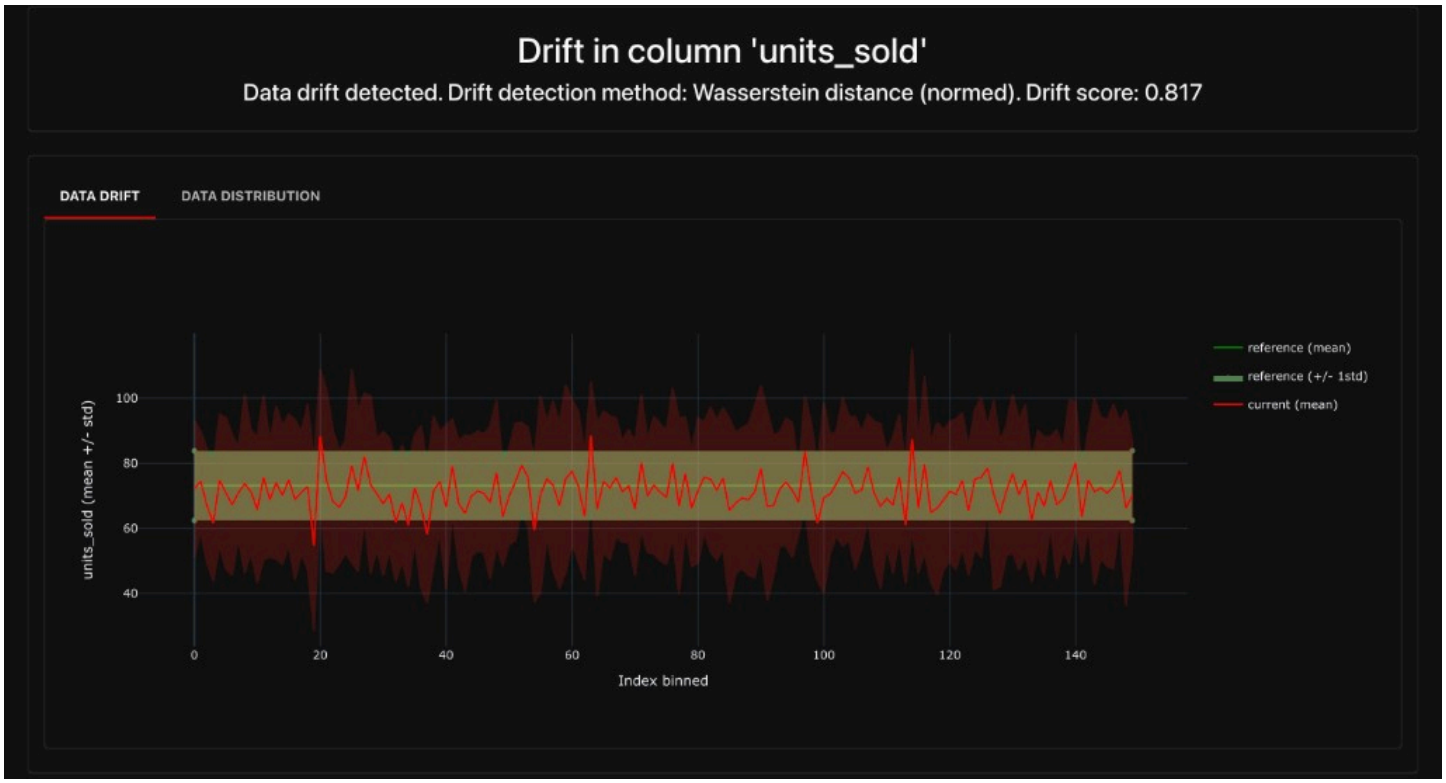
1. Prefect Flow Success - drift-triggered retrain completed



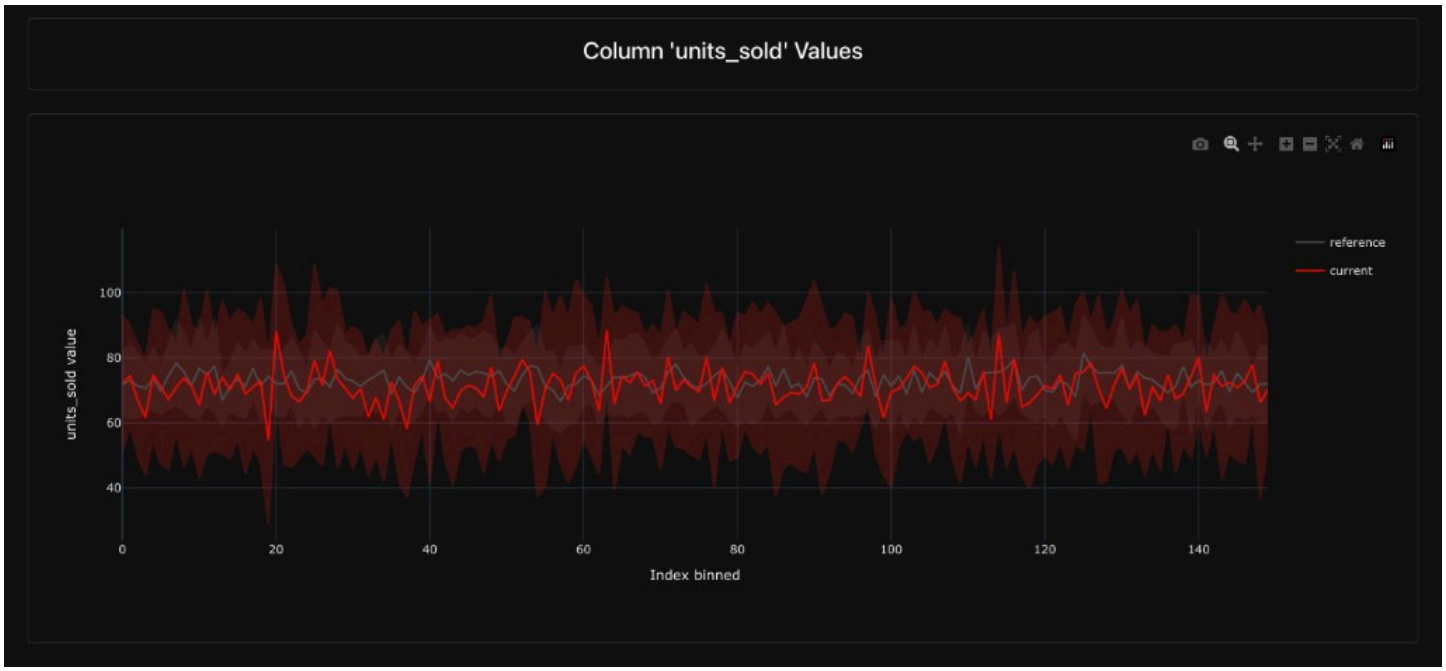
2. Dataset Drift Summary (Evidently AI)



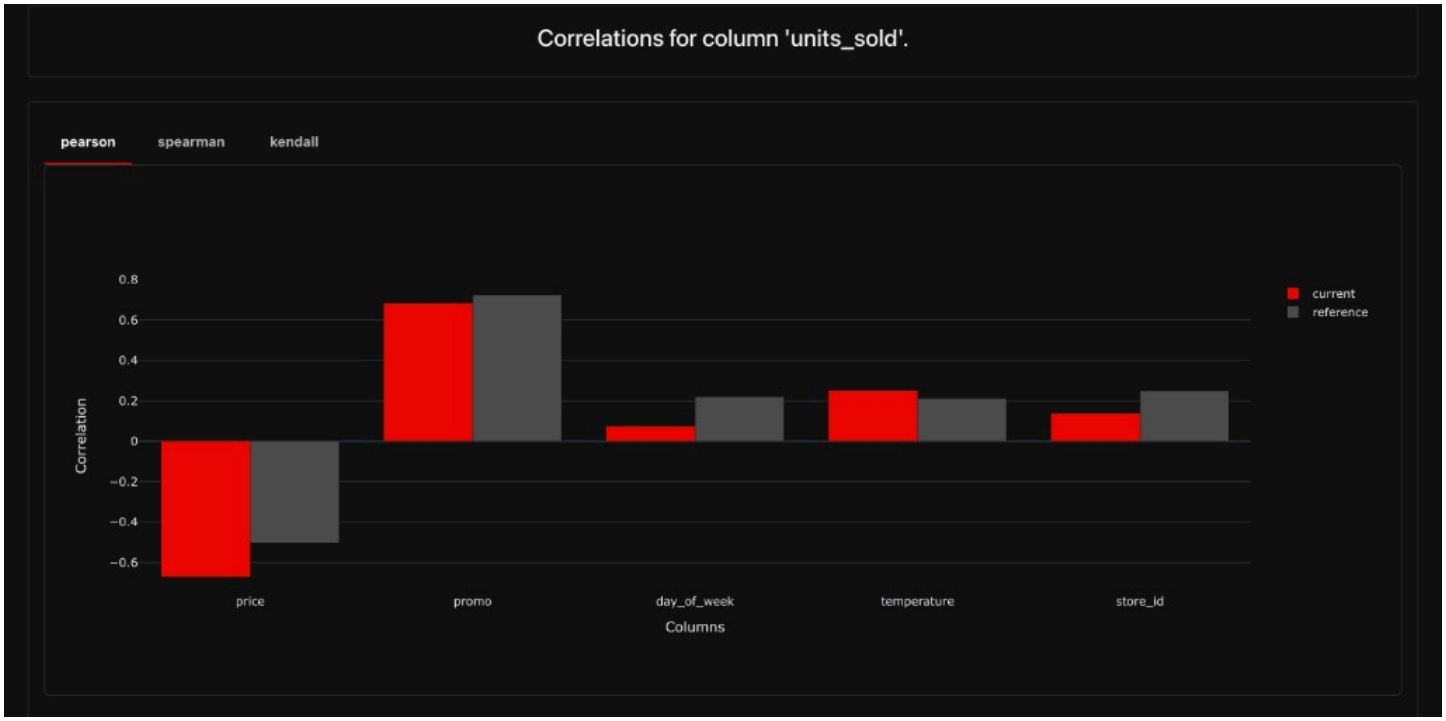
2. Drift in column units_sold (DATA DRIFT)



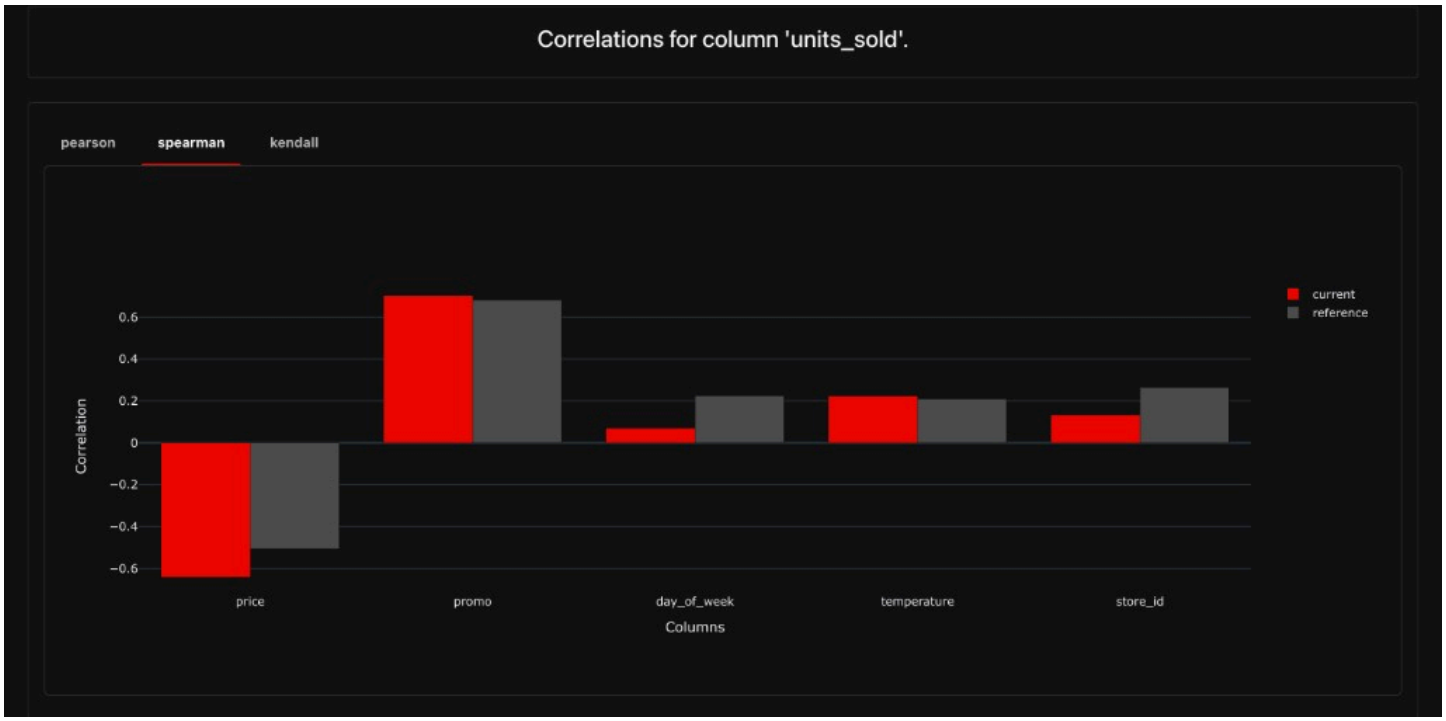
2. Column 'units_sold' Values



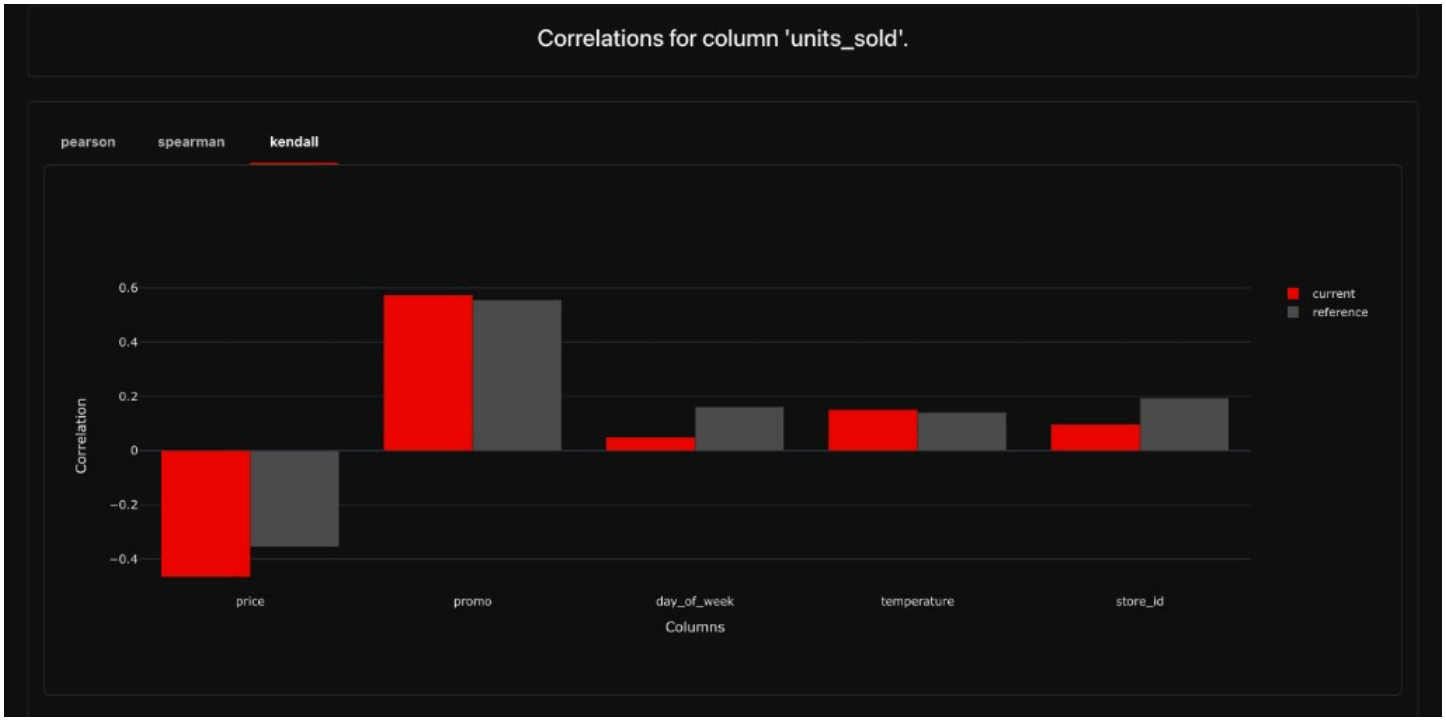
2. Correlations (Pearson) for units_sold



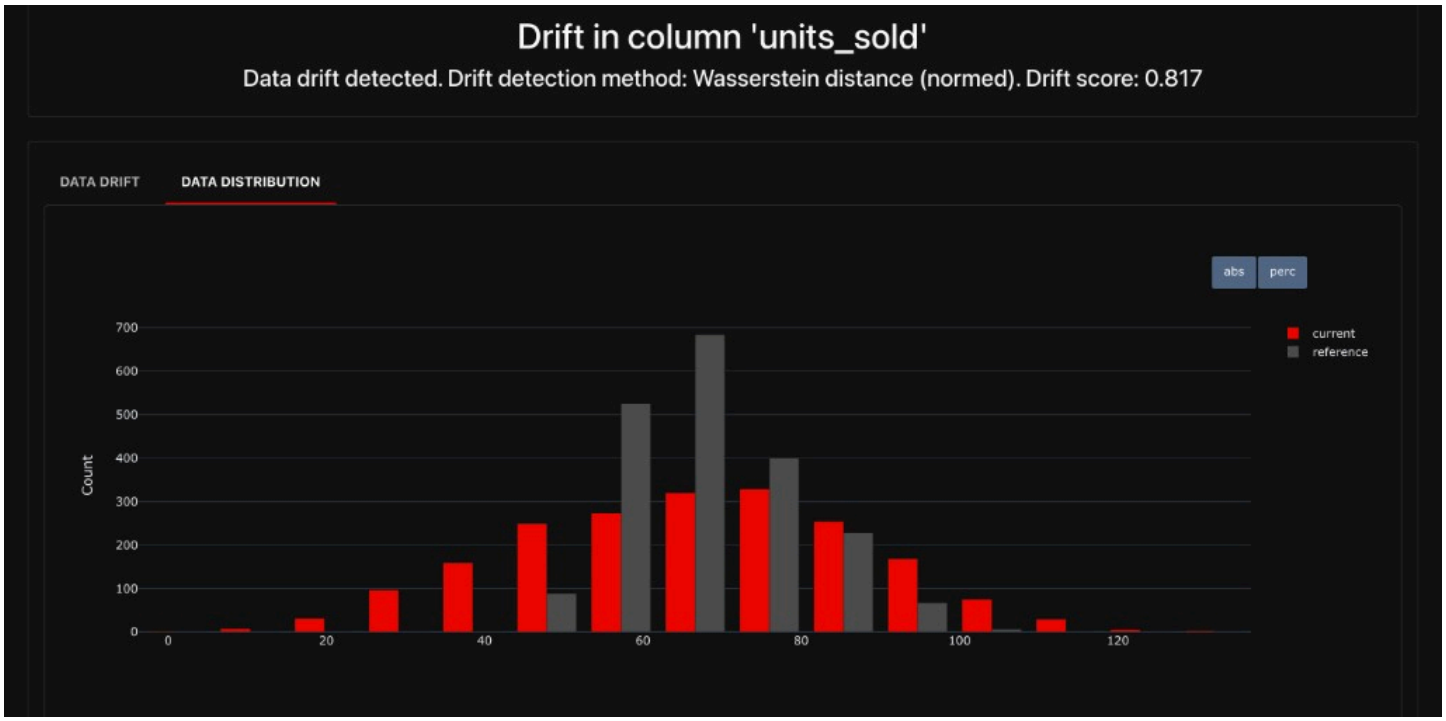
2. Correlations (Spearman) for units_sold



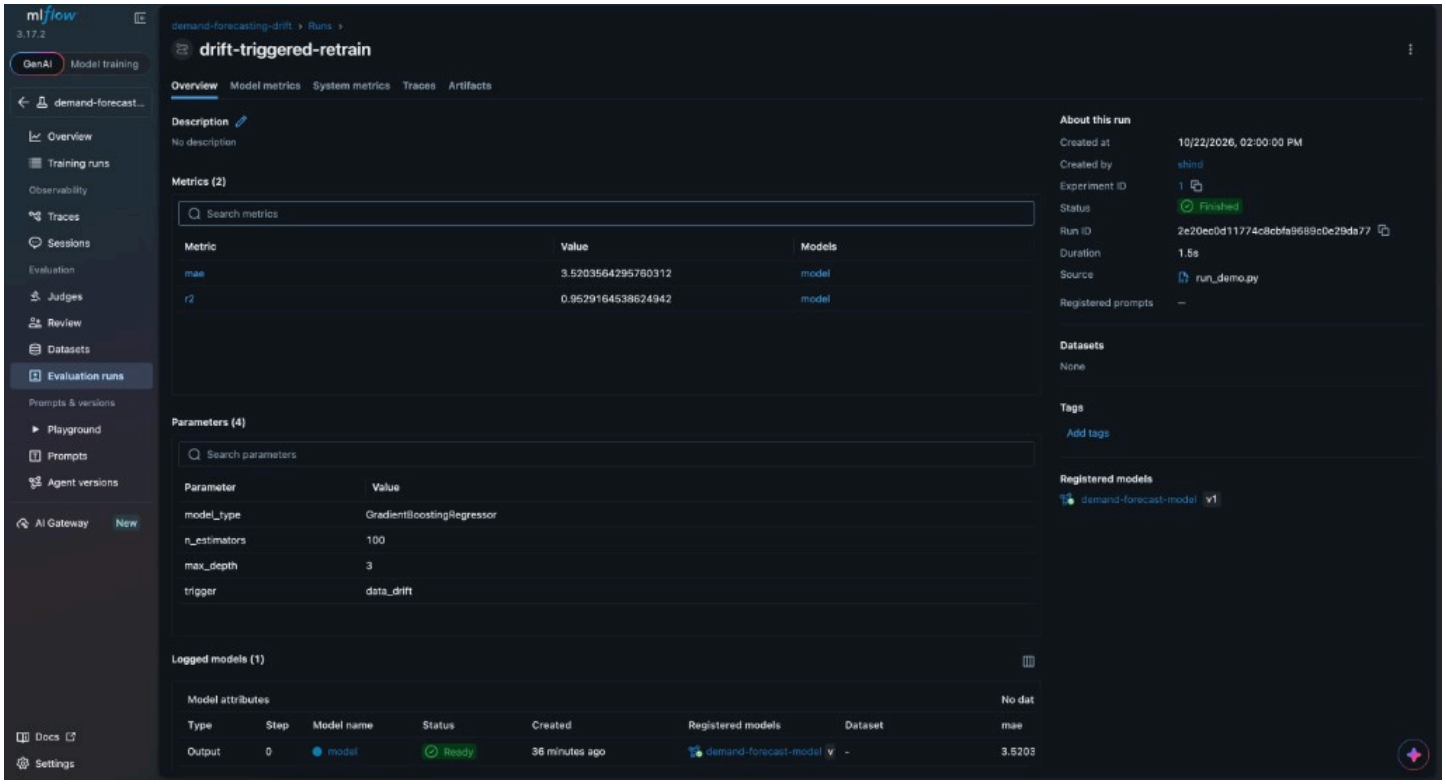
2. Correlations (Kendall) for units_sold



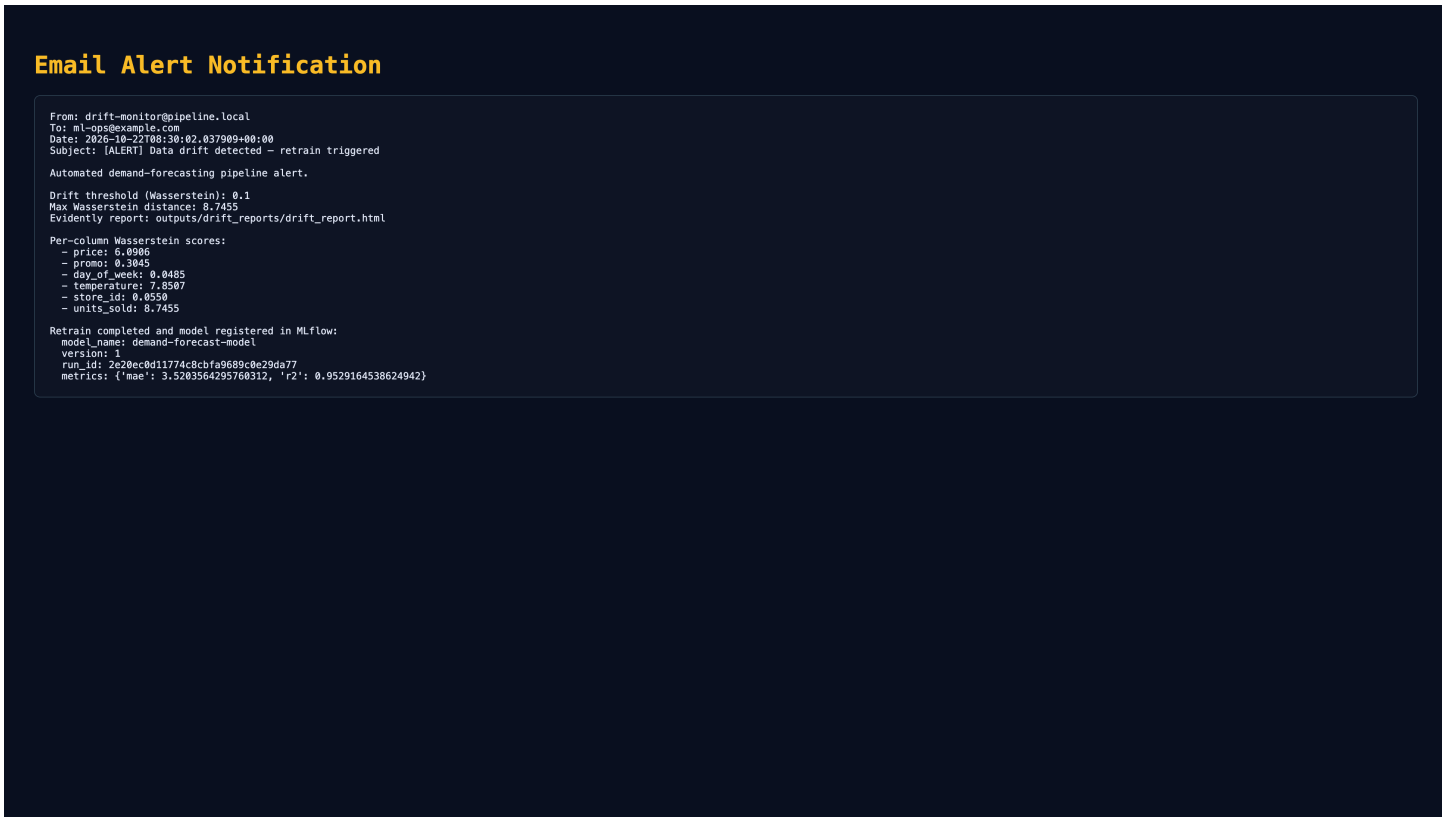
2. units_sold DATA DISTRIBUTION



3. MLflow - drift-triggered-retrain + demand-forecast-model v1



4. Email Alert Notification - retrain triggered



5. GitHub Repository - ADT23SOCB1617-drift-retraining-pipeline

